



Extended Calculus Topic Test 1

Solution

Total Marks: 30

Time: 50 minutes

1 Differentiate the following with respect to x :

(i) $\frac{\cos 3x}{x^2}$

$$\begin{aligned} u &= \cos 3x & v &= x^2 \\ u' &= -3 \sin 3x & v' &= 2x \\ y' &= \frac{-3x^2 \sin 3x - 2x \cos 3x}{x^4} \\ &= \frac{-x(3x \sin 3x + 2 \cos 3x)}{x^4} \\ &= \frac{-(3x \sin 3x + 2 \cos 3x)}{x^3} \end{aligned}$$

2

(ii) $\ln \sqrt{3-x}$

$$\begin{aligned} &= \ln (3-x)^{\frac{1}{2}} = \frac{1}{2} \ln (3-x) \\ y' &= \frac{1}{2} \left(\frac{-1}{3-x} \right) \\ &= \frac{-1}{2(3-x)} = \frac{1}{2(x-3)} \end{aligned}$$

2

2 Differentiate the following with respect to x :

(i) $\frac{e^{x^2}}{2x+3}$

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{(2x+3)2xe^{x^2} - e^{x^2} \cdot 2}{(2x+3)^2} = \frac{2e^{x^2} [2x^2 + 3x - 1]}{(2x+3)^2}$$

2

(ii) $\ln \left(\frac{1+x}{1-x} \right)$

$$\begin{aligned} y &= \ln \left(\frac{1+x}{1-x} \right) = \ln(1+x) - \ln(1-x) \\ \frac{dy}{dx} &= \frac{1}{1+x} - \frac{1}{1-x} \\ &= \frac{1-x - (1+x)}{1-x^2} = \frac{-2x}{1-x^2} \end{aligned}$$

2

3 Differentiate $x^3 e^{2x}$ with respect to x .

$$\frac{d}{dx} x^3 e^{2x} = x^3 \times 2e^{2x} + e^{2x} \times 3x^2 = x^2 e^{2x} (2x+3)$$

2

4 Differentiate $e^{\cos(\pi x)}$.

$$\frac{d}{dx} x^3 e^{2x} = x^3 \times 2e^{2x} + e^{2x} \times 3x^2 = x^2 e^{2x} (2x+3)$$

2

5 Given that the function $f(x)$ has a derivative $y' = 4e^{4x} + 3$ and the equation of the tangent to this curve is $y = 7x + 2$. Find the exact value of $f(3)$.

3

$$\begin{aligned} \text{m.tang} &= 7 \\ 7 &= 4e^{4x} + 3 \\ 4 &= 4e^{4x} \\ 1 &= e^{4x} \\ e^0 &= e^{4x} \\ x &= 0 \\ y &= 7(0) + 2 = 2 \end{aligned}$$

$$\begin{aligned} y &= \int (4e^{4x} + 3) dx \\ y &= e^{4x} + 3x + C \\ 2 &= e^0 + 3(0) + C \\ C &= 1 \\ f(3) &= e^{12} + 12 + 1 \\ f(3) &= e^{12} + 13 \end{aligned}$$

6 If $y = \log_e x$, show that $\frac{d^2 y}{dx^2} + \left(\frac{dy}{dx} \right)^2 = 0$.

2

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{x} \quad \text{and} \quad \frac{d^2 y}{dx^2} = -\frac{1}{x^2} \\ \text{LHS} &= \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx} \right)^2 = -\frac{1}{x^2} + \left(\frac{1}{x} \right)^2 = -\frac{1}{x^2} + \frac{1}{x^2} = 0 \\ &= \text{RHS, as required} \end{aligned}$$

7 Find $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot x \, dx$.

$$\begin{aligned} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot x \, dx &= \left[\ln(\sin x) \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} \\ &= \ln\left(\sin \frac{\pi}{2}\right) - \ln\left(\sin \frac{\pi}{4}\right) \\ &= \ln 1 - \ln\left(\frac{1}{\sqrt{2}}\right) \\ &= 0 - \ln 2^{-\frac{1}{2}} = \frac{1}{2} \ln 2 \end{aligned}$$

3

8 Find:

(i) $\int \frac{4 \sin\left(\frac{5x}{3}\right)}{7} \, dx$

$$\begin{aligned} &\frac{4}{7} \int \sin\left(\frac{5x}{3}\right) \, dx \\ &= \frac{12}{35} \left[-\cos\left(\frac{5x}{3}\right) \right] + C \\ &= -\frac{12}{35} \cos\left(\frac{5x}{3}\right) + C \end{aligned}$$

2

(ii) $\int \sec^2 5x \, dx = \frac{1}{5} \tan 5x + C$

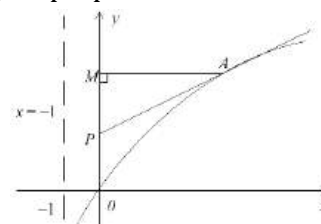
1

9 Find $\int_{-2}^1 \frac{2}{x+3} \, dx$.

$$2 \int_{-2}^1 \frac{1}{x+3} \, dx = 2 \left[\ln(x+3) \right]_{-2}^1 = 2(\ln 4 - \ln 1) = 2 \ln 4$$

2

10 In the diagram, the point A whose y-coordinate is 1, lies on the curve $y = \ln(x+1)$. The tangent to the curve at A cuts the y-axis at P. A straight line through A perpendicular to the y-axis meets the y-axis at M.



(i) Show that the x-coordinate of A is $(e-1)$.

$$\begin{aligned} \text{To find A let } y=1 \text{ in } y &= \ln(x+1) \\ \therefore 1 &= \ln(x+1) \\ x+1 &= e \quad x = e-1 \end{aligned}$$

1

(ii) Show that the equation of the tangent AP is $x - ey + 1 = 0$.

4

$$\begin{aligned} y &= \ln(x+1) \quad \frac{dy}{dx} = \frac{1}{x+1} \\ \text{When } x &= e-1 \quad \frac{dy}{dx} = \frac{1}{e} \\ \text{For the tangent AP } y-1 &= \frac{1}{e}(x-(e-1)) \\ ey - e &= x - e + 1 \\ \therefore x - ey + 1 &= 0 \end{aligned}$$



Extended Calculus Topic Test 2

Solution

Total Marks: 30

Time: 50 minutes

1 (i) Differentiate $y = \frac{1}{2} \ln(x^2)$ with respect to x . $y' = \frac{1}{2} \times \frac{2x}{x^2} = \frac{x}{x^2} = \frac{1}{x}$ 2

(ii) Differentiate $y = \frac{e^x}{\sin x}$ with respect to x . $y' = \frac{\sin x \times e^x - e^x (\cos x)}{\sin^2 x} = \frac{e^x (\sin x - \cos x)}{\sin^2 x}$ 2

2 Differentiate with respect to x :

(i) $\log_e \left(\frac{x}{x+1} \right)$ $\frac{d}{dx} \log_e \left(\frac{x}{x+1} \right) = \frac{d}{dx} [\log_e x - \log_e (x+1)] = \frac{1}{x} - \frac{1}{x+1}$ 2

(ii) $\frac{\sin x}{3x-4}$ $\frac{d}{dx} \left(\frac{\sin x}{3x-4} \right) = \frac{\cos x (3x-4) - 3 \sin x}{(3x-4)^2}$ 2

3 Find derivatives for the following, with respect to x .

(i) $\ln(x^2 + 2)$ $\frac{2x}{x^2 + 2}$ $\frac{d}{dx} e^{x \ln 3} = e^{x \ln 3} \ln 3 = 3^x \ln 3$ 1

(ii) $3^x e^x$ $\frac{d}{dx} 3^x e^x = 3^x e^x + 3^x \ln 3 = 3^x e^x (1 + \ln 3)$ 2

4 (i) Differentiate $x^2 \sin 2x$. $u = x^2$ $v = \sin 2x$
 $u' = 2x$ $v' = 2 \cos 2x$
 $y' = 2x^2 \cos 2x + 2x \sin 2x$ 2

(ii) Hence or otherwise, evaluate $\int_0^{\frac{\pi}{4}} (x^2 \cos 2x + x \sin 2x) dx$. $= \frac{1}{2} [x^2 \sin 2x]_0^{\frac{\pi}{4}}$
 $= \frac{1}{2} \times \frac{\pi^2}{16} \times 1 = \frac{\pi^2}{32}$ 2

5 (i) Show that $\frac{d}{dx} (x \cos x - \sin x) = -x \sin x$. $\frac{d}{dx} (x \cos x - \sin x) = \sin x - x \sin x - \cos x = -x \sin x$, as required. 1

(ii). Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{4}} x \sin x dx$. $\int_0^{\frac{\pi}{4}} x \sin x dx = -[x \cos x - \sin x]_0^{\frac{\pi}{4}} = -\left(\frac{\pi}{4} \times \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}} \left(1 - \frac{\pi}{4}\right) = \frac{1}{4\sqrt{2}} (4 - \pi)$ 2

6 (i) Find $\frac{d}{dx} \log_e (1 + 2 \tan x)$. $\frac{d}{dx} \log_e (1 + 2 \tan x) = \frac{2 \sec^2 x}{1 + 2 \tan x}$ 1

(ii) Hence evaluate $\int_0^{\frac{\pi}{4}} \frac{\sec^2 x}{1 + 2 \tan x} dx$, giving your answer in exact form.
 $\int_0^{\frac{\pi}{4}} \frac{\sec^2 x}{1 + 2 \tan x} dx = \frac{1}{2} \int_0^{\frac{\pi}{4}} \frac{2 \sec^2 x}{1 + 2 \tan x} dx$
 $= \frac{1}{2} [\log_e (1 + 2 \tan x)]_0^{\frac{\pi}{4}}$
 $= \frac{1}{2} \left[\log_e \left(1 + 2 \tan \frac{\pi}{4}\right) - \log_e (1 + 2 \tan 0) \right] = \frac{1}{2} \log_e 3$ 2

7 The curve $y = f(x)$ passes through the point $(0, 7)$. If its gradient is given by $\frac{dy}{dx} = 1 - 6 \sin 3x$, 2

find the equation of the curve.
 $\frac{dy}{dx} = 1 - 6 \sin 3x$
 $f(x) = x + \frac{6 \cos 3x}{3} + C$
 $f(0) = 0 + \frac{6 \cos 0}{3} + C = 7$
 $7 = 2 + C \therefore C = 5$
 $f(x) = x + 2 \cos 3x + 5$

8 (i) Differentiate $\sin^2 x$. $\frac{dy}{dx} = 2 \cos x \sin x$ 1

(ii) Hence, calculate $\int_0^{\frac{\pi}{4}} (\sin x + \cos x)^2 dx$, leaving your answer in exact form. 2

$$\begin{aligned} &= \int_0^{\frac{\pi}{4}} (\sin^2 x + 2 \sin x \cos x + \cos^2 x) dx \\ &= \int_0^{\frac{\pi}{4}} (1 + 2 \sin x \cos x) dx = \left[x + \sin^2 x \right]_0^{\frac{\pi}{4}} \\ &= \left[\frac{\pi}{4} + \left(\frac{1}{\sqrt{2}} \right)^2 \right] - 0 = \frac{\pi}{4} + \frac{1}{2} \end{aligned}$$

9 Find the value of k such that $\int_{-2}^0 \frac{x^2}{x^3-2} dx = \ln k$. $\ln k = \frac{1}{3} \int_{-2}^0 \frac{3x^2}{x^3-2} dx = \frac{1}{3} \left[\ln |x^3-2| \right]_{-2}^0$ 2

$$\begin{aligned} &= \frac{1}{3} (\ln 2 - \ln 10) = \frac{1}{3} \ln \left(\frac{1}{5} \right) \\ \therefore k &= \sqrt[3]{\frac{1}{5}} \end{aligned}$$

10 Find the exact gradient of the tangent to the curve $y = x \tan x$ at the point where $x = \frac{\pi}{3}$. 2

$$\begin{aligned} y &= x \tan(x) \\ \frac{dy}{dx} &= 1 \times \tan(x) + x \times \sec^2(x) \\ &= \tan(x) + x \sec^2(x) \\ \text{At } x &= \frac{\pi}{3}, \frac{dy}{dx} = \sqrt{3} + \frac{4\pi}{3} \end{aligned}$$